

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2284

Salmon Creek 138 kV
November 23, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2284. The Interconnection Customer proposes a 91.7 MW solar photovoltaic facility interconnecting at the Salmon Creek 138 kV substation in the Pennsylvania area. The request was received on January 27, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 770 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,424,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: LARNED – DEPOE BAY 138kV
Contingency: FLORENCE MILL – GADSDEN 138kV L
Loading: 122.0% (Rating: 215 MVA)
DFAX: 56.5%
Pre-project loading: 97.9%
2. Facility: SALMON CRK – DEPOE BAY 138kV
Contingency: FLORENCE MILL – GADSDEN 138kV L
Loading: 114.6% (Rating: 330 MVA)
DFAX: 100.0%
Pre-project loading: 86.8%
3. Facility: FLORENCE MILL – GADSDEN 138kV
Contingency: SALMON CREEK – DEPOE BAY 138kV
Loading: 114.6% (Rating: 330 MVA)
DFAX: 100.0%
Pre-project loading: 86.8%
4. Facility: SALMON CREEK – GADSDEN 138kV
Contingency: SALMON CRK SUBSTATION – DEPOE BAY 138kV
Loading: 112.8% (Rating: 212 MVA)
DFAX: 100.0%
Pre-project loading: 69.6%
5. Facility: TEMPLE SWITCHING – LARNED 138kV
Contingency: FLORENCE MILL – GADSDEN 138kV
Loading: 110.0% (Rating: 233 MVA)
DFAX: 56.5%
Pre-project loading: 87.8%

6. Facility: MCADORY – THERMO SUBSTATION 138kV
Contingency: WAHIAWA – SUNRISE TR 345kV L
Loading: 104.8% (Rating: 206 MVA)
DFAX: 6.2%
Pre-project loading: 102.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LARNED – DEPOE BAY 138kV	Rebuild line with higher-capacity conductor	\$4,098,000	17% (\$695,000)
DEPOE BAY – SALMON CRK 138kV L	Rebuild line with higher-capacity conductor	\$16,764,000	17% (\$2,845,000)
FLORENCE MILL – GADSDEN 138kV Line	Rebuild line with higher-capacity conductor	\$3,533,000	17% (\$600,000)
SALMON CREEK – GADSDEN 138kV	Rebuild line with higher-capacity conductor	\$4,748,000	23% (\$1,095,000)
TEMPLE SWITCHING – LARNED 138kV line	Rebuild line with higher-capacity conductor	\$4,190,000	17% (\$711,000)
MCADORY – THERMO SUBSTATION 138kV	Reconductor line	\$3,653,000	13% (\$478,000)
	Total allocated to project		\$6,424,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.